Biology — Exam 02 (Chapter 7: Metabolism)

Federal Board of Intermediate and Secondary Education, Islamabad | Class IX Paper: SSC-I
Attempt date: 2026-05-02 | Marked: 2026-05-02 | Syllabus: Chapter 7 — Metabolism (Sections 7.1–7.8) | RE-SIT (Previous attempt: 2026-04-30)

PERFECT SCORE — 65/65 (100%)   Second Perfect Score Overall   Consecutive MCQ Perfection Streak: 8

All three weaknesses identified in the April 30 marking report have been explicitly corrected in this attempt.

65/65	100%	12/12	33/33	20/20	A+
TOTAL MARKS	PERCENTAGE	SECTION A MCQS	SECTION B SHORT	SECTION C LONG	GRADE

SECTION A — Multiple Choice Questions   12 × 1 = 12 Marks | Seemab: 12/12

#	Question (abbreviated)	A	B	C	D	Seemab's Answer	Correct	Result
1	Sum of all chemical reactions in living cell	Catabolism	Anabolism	Metabolism	Enzymology	**C**	C	1/1
2	Enzymes are biologically active globular proteins produced by	Dead cells	Living cells	Inorganic ions	Ribosomes only	**B**	B	1/1
3	Part of enzyme where substrate binds and reaction occurs	Allosteric site	Prosthetic group	Active site	Coenzyme	**C**	C	1/1
4	At very high temperature, enzyme loses function because its structure is	Activated	Denatured	Phosphorylated	Synthesised	**B**	B	1/1
5	In competitive inhibition, inhibitor blocks enzyme by binding to the	Allosteric site	Substrate	Active site	Cofactor	**C**	C	1/1
6	ATP is called "energy currency" of the cell because it	Is in nucleus	Stores and releases energy	Builds proteins	Carries oxygen	**B**	B	1/1
7	ATP molecule consists of adenine, ribose sugar and	Two phosphate groups	Three phosphate groups	One phosphate group	Four phosphate groups	**B**	B	1/1
8	During photosynthesis, oxygen is released as a by-product through	Calvin cycle	Photolysis of water	Glycolysis	Krebs cycle	**B**	B	1/1
9	Light-independent reactions (Calvin cycle) occur in the	Thylakoid membrane	Cytoplasm	Stroma of chloroplast	Mitochondrial matrix	**C**	C	1/1
10	Glycolysis produces 2 pyruvic acid from glucose with net gain of	36 ATP	4 ATP	2 ATP	38 ATP	**C**	C	1/1
11	Anaerobic respiration in yeast (alcoholic fermentation) produces ethyl alcohol and	Water	Lactic acid	Carbon dioxide	Oxygen	**C**	C	1/1
12	Final electron acceptor in ETC is	NADH	Carbon dioxide	Oxygen	Water	**C**	C	1/1

SECTION B — Short Questions   11 × 3 = 33 Marks | Seemab: 33/33 (All 11 attempted)

Q.2(i) OR — Why enzymes are biological catalysts; two characteristics **3/3**

Why are enzymes called biological catalysts? Give two characteristics to support this.

✓ Enzymes as biological catalysts defined — made of specific proteins in the body, biologically active globular proteins made by living cells that tremendously speed up biochemical reactions. **1 mark.**

✓ Characteristic 1: Increase rate of reaction — clearly stated. **1 mark.**

✓ Characteristic 2: Required in small quantity — "one enzyme catalyses a huge amount of substrate" (equivalent to: enzyme regains shape and is reused). **1 mark.**

Full marks. Both characteristics clearly stated with supporting detail. Clean answer.

Q.2(ii) — Name and describe three types of cofactors with one example each **3/3**

Name and describe three types of cofactors with one example each.

✓ Activators — detachable inorganic ions; examples: Zn^{++}, Fe^{++}, Cu^{++} and chloride ions (salivary amylase activity increased by chloride ions). **1 mark.**

✓ Prosthetic group — organic molecule tightly and permanently attached to the enzyme; examples: FAD (Flavin Adenine Dinucleotide), Haem group. **1 mark.**

✓ Coenzymes — detachable organic molecule; examples: NAD (nicotinamide adenine dinucleotide), Coenzyme A, vitamins A and C. **1 mark.**

Perfect — all three named, described and exemplified correctly. Note that the answer key lists vitamins A and C as coenzyme examples — Seemab correctly included these.

Q.2(iii) OR — Mechanism of enzyme action (ES and EP complex) **3/3**

What is energy of activation? How do enzymes lower activation energy?

✓ Step 1: Specific substrate binds to the active site → forms Enzyme-Substrate (ES) complex. **1 mark.**

✓ Step 2: Enzyme catalyses the reaction, converting substrate to product → forms EP complex (Enzyme-Product complex). **1 mark.**

✓ Step 3: EP complex breaks up; products are released; enzyme is unchanged and available again for the next reaction. **1 mark.**

All three steps stated correctly with correct complex names. Clean three-step structure matches marking scheme perfectly.

Q.2(iv) — Effect of temperature on enzyme activity **3/3**

How does temperature affect enzyme activity? What happens at very high and very low temperatures?

✓ Optimum temperature: Each enzyme performs best at specific temperature. For human enzymes: 36–38°C. Rate increases with temperature up to this point. **1 mark.**

✓ Very high temperature: Enzyme atoms vibrate vigorously → globular structure and active site damaged → denaturation (permanent loss of activity). **1 mark.**

✓ Very low temperature: Insufficient activation energy → enzyme becomes inactive (reversible — activity resumes when temperature is restored). **1 mark.**

All three temperature zones covered with correct mechanism and reversibility distinction. "Permanent" for denaturation explicitly stated.

Q.2(v) OR — Allosteric site; how non-competitive inhibitor damages active site **3/3**

What is the allosteric site? How does a non-competitive inhibitor damage the active site?

✓ Allosteric site: "A point on the enzyme surface other than the active site where non-competitive inhibitors attach." Named explicitly and correctly. **1 mark.**

✓ Mechanism: Inhibitor binding alters the globular shape of the enzyme → deforms and damages the active site. **1 mark.**

✓ Result: Substrate cannot bind to the damaged active site; no ES complex forms; reaction stops. **1 mark.**

IMPROVEMENT NOTED: In the April 30 attempt, Seemab wrote about non-competitive inhibition but did not explicitly name the allosteric site as the binding location — losing 1 mark. This time the allosteric site is defined precisely and named at the start of the answer. Weakness fully corrected.

Q.2(vi) OR — ATP-ADP cycle; energy storage and release **3/3**

Explain the ATP-ADP cycle. How is energy stored and released in this cycle?

✓ Energy storage: ADP + inorganic phosphate (P_i) combine by condensation to reform ATP. Energy from catabolism (or light in plants) drives this — stored in phosphate bond. **1 mark.**

✓ Energy release: ATP is hydrolysed — third phosphate separates → releases energy + ADP + P_i. **1 mark.**

✓ Significance: Energy available on demand for movement, active transport, growth, nerve impulse, gland secretion — without wasteful overproduction. **1 mark.**

Condensation and hydrolysis terms used correctly. Significance paragraph matches answer key precisely. Full marks.

Q.2(vii) OR — Three limiting factors of photosynthesis — 3/3

Name three limiting factors of photosynthesis and briefly explain how each affects the rate.

✓ Light intensity: More light intensity and longer duration increase photosynthesis rate by providing more energy for light reaction. **1 mark.**

✓ CO_2 concentration: CO_2 is raw material (substrate) for the photosynthetic reaction. More CO_2 increases rate, but only up to a specific limit. **1 mark.**

✓ Temperature: Optimum temperature for photosynthesis is 25°C. Temperature below or above the optimum decreases rate by affecting enzyme activity. **1 mark.**

All three factors with correct mechanisms. The "specific limit" for CO_2 shows understanding of enzyme saturation. Optimum temperature value (25°C) correctly stated for photosynthesis.

Q.2(viii) — Light-dependent reactions: location and products — 3/3

What are light-dependent reactions of photosynthesis? Where do they occur and what are the products?

✓ Definition: Reactions that require light energy directly; they depend on Photosystems I and II to absorb light and begin photosynthesis. **1 mark.**

✓ Location: Thylakoid membrane (grana) of chloroplast, where photosynthetic pigments are arranged into photosystems. **1 mark.**

✓ Products: ATP, NADPH and oxygen (from photolysis of water). ATP and NADPH power the Calvin cycle. **1 mark.**

All three components present. "Grana" correctly identified as the structural unit. Products complete including the role of ATP/NADPH in powering the Calvin cycle.

Q.2(ix) OR — Aerobic respiration equation; ATP yield per glucose — 3/3

Write the equation for aerobic respiration and state how many ATP molecules are produced from one glucose molecule.

✓ Equation: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy (ATP). Correctly balanced. **1 mark (ATP yield).**

✓ ATP yield: Complete aerobic respiration yields 36 ATP per glucose (2 glycolysis + 2 Krebs + 32 ETC). Breakdown correct. **1 mark (comparison).**

✓ Requirement: Requires oxygen as final electron acceptor; without O_2 only glycolysis + fermentation occurs. **1 mark (requirement).**

IMPROVEMENT NOTED: In April 30, Seemab omitted the "inner mitochondrial membrane" location for ETC in Q2(ix) main question, losing 1 mark. This time the OR was chosen and all three marks are covered cleanly — 32 ATP from ETC breakdown shown, oxygen requirement stated explicitly. Full correction achieved.

Q.2(x) OR — Role of the Krebs cycle; products formed — 3/3

What is the role of the Krebs cycle in aerobic respiration? Name the products formed.

✓ Role: Cyclic process in mitochondrial matrix; acetyl CoA is fully oxidised; energy extracted as electron carriers (NADH) that feed the ETC. **1 mark.**

✓ Products: $4CO_2 + 2ATP + 6NADH + 2FADH_2$ per glucose. Both NADH and $FADH_2$ named. **1 mark.**

✓ Significance: NADH and $FADH_2$ carry electrons to ETC where most ATP is produced; CO_2 released as waste. **1 mark.**

Mitochondrial matrix location correctly stated. Products correctly broken down with both electron carriers named. Significance links correctly to ETC.

Q.2(xi) OR — Comparison: alcoholic fermentation vs lactic acid fermentation — 3/3

Compare alcoholic fermentation and lactic acid fermentation. Give one biological example of each.

✓ Organisms: Yeast, some bacteria (alcoholic) vs Muscle cells, lactic acid bacteria (lactic acid). **(contributes to product mark)**

✓ Products: Ethyl alcohol + CO_2 vs Lactic acid only. **1 mark.**

✓ Equation: Pyruvic acid → Ethyl alcohol + CO_2 vs Pyruvic acid → Lactic acid. **1 mark.**

✓ Example: Wine making vs Yogurt/cheese. **1 mark.**

Clear comparison table format. All required elements: organisms, products, equations and examples — all correctly matched. Clean presentation.

SECTION C — Long Questions 4 × 5 = 20 Marks | Seemab: 20/20 (All 4 attempted)

Q.3(i) OR — Three environmental factors affecting enzyme activity (temp, pH, substrate conc) 5/5

Explain how temperature, pH, and substrate concentration affect enzyme activity. Include denaturation and saturation.

✓ Temperature (1 mark): Optimum 36–38°C for humans. Above optimum: enzyme atoms vibrate, globular structure damaged → denaturation (permanent). Below optimum: enzyme inactive (reversible). Rate rises with temperature to optimum.

✓ pH (1 mark): Each enzyme has specific optimum pH. Change effects ionisation of amino acids at active site. Pepsin pH 2.00 (stomach); Trypsin pH 8.00 (small intestine). Extreme pH → denaturation.

✓ Substrate concentration (2 marks): Rate directly proportional to substrate concentration when enzyme molecules are sufficient. Rate increases until saturation of active sites — all sites occupied. Beyond saturation more substrate has no effect. This plateau is the maximum rate (Vmax). Both saturation concept AND Vmax named.

✓ Inhibitors and activators (1 mark): Inhibitors (poisons, cyanide, antibiotics, drugs) reduce/stop enzyme activity. Activators increase rate. Both regulate enzymes from inside or outside the cell.

Outstanding answer covering all five marking points: temperature (1), pH (1), substrate concentration with saturation + Vmax (2), and inhibitors/activators (1). Vmax explicitly named as "maximum rate (Umax)" — full 2-mark substrate concentration credit awarded. Pepsin and Trypsin pH values correctly stated.

Q.3(ii) — Mechanism of enzyme action; activation energy; lock-and-key vs induced-fit 5/5

Describe the mechanism of enzyme action. Explain how an enzyme lowers activation energy and distinguish the lock-and-key concept from the induced-fit model.

✓ Active site (1 mark): "Charge-bearing binds to an active site — very small, specific shape. Substrate binds here and reaction occurs." Correctly identifies active site as charge-bearing and specific shape.

✓ Mechanism — 3 steps (1 mark): (1) Substrate binds active site → ES complex; (2) Enzyme catalyses reactions → EP complex; (3) Products released → enzymes unchanged. All three steps present.

✓ Lowering activation energy (1 mark): "Enzymes change shape/charge of substrate at active site or provides a different reaction mechanism → reactions proceed at body temperature without extreme conditions." Correct.

✓ Lock-and-Key model (1 mark): Emil Fischer. Active site is rigid and fixed (lock); substrate is a specific key. Explains why only one enzyme fits only one specific substrate.

✓ Induced-Fit model (1 mark): More accepted. Active site is flexible. When the correct substrate approaches, active site changes shape to accommodate it precisely. The site is still specific — only the correct substrate can induce the correct conformational change.

All five marks covered in excellent detail. Emil Fischer credited for Lock-and-Key. Induced-Fit correctly described as "more accepted" with conformational change explained. The contrast between rigid (Lock-and-Key) and flexible (Induced-Fit) is clearly drawn.

Q.3(iii) — ATP as energy currency; structure; ATP-ADP cycle; uses in the body 5/5

Explain the role of ATP as the energy currency of the cell. Describe the structure of ATP, the ATP-ADP cycle, and where ATP is used in the body.

✓ ATP as energy currency (1 mark): "Living organisms use and store energy at cellular level in the form of Adenosine Triphosphate (ATP). Called energy currency because it stores and releases energy like money — converted between forms as needed. Human cells use 100–150 moles of ATP per day." Precise and complete.

✓ Structure of ATP (1 mark): Three components: (1) Adenine (double-ringed nitrogen base); (2) Ribose sugar (5C); (3) Three phosphate groups (PO_4) in triphosphate chain. Bonds between phosphate groups = high-energy bonds (wavy lines). AMP = 1 phosphate; ADP = 2 phosphates; ATP = 3 phosphates.

✓ ATP-ADP cycle (2 marks): Diagram drawn — energy from light (plants) → ADP + P_i by condensation → ATP; ATP hydrolysis → energy for cellular work → ADP + P_i. Both condensation (for storage) and hydrolysis (for release) correctly shown in cycle.

✓ Uses of ATP (1 mark): Working of muscles, nerve impulse transmission, growth and repair of cells, active transport, bilandi/gland secretion. Five uses listed.

Complete and precise. The 100–150 moles/day figure correctly cited. Structural components fully named. The ATP-ADP cycle drawn as a diagram with condensation/hydrolysis labelled — exactly what the answer key requires for the 2-mark component.

Q.3(iv) OR — Aerobic respiration: four stages with ATP yield at each stage 5/5

Describe the mechanism of aerobic respiration in four stages: glycolysis, formation of Acetyl CoA, Krebs cycle, and ETC. State the ATP yield at each stage.

✓ Stage 1 — Glycolysis (1 mark): Cytoplasm. Glucose → 2 pyruvic acid. Net yield: 2ATP + 2NADH. Common to aerobic and anaerobic.

✓ Stage 2 — Formation of Acetyl CoA (1 mark): Mitochondria. Pyruvic acid oxidised; 2C acetyl group + CO_2; CoA → Acetyl CoA. Equation: 2 pyruvic acid + 2CoA → 2 Acetyl CoA + $2CO_2$ + 2NADH. Yield: 2 NADH per glucose.

✓ Stage 3 — Krebs Cycle (1 mark): Mitochondrial matrix. Acetyl group fully oxidised. Yield: $4CO_2$ + 2ATP + 6NADH + $2FADH_2$ per glucose. Both NADH and $FADH_2$ named with correct quantities.

✓ Stage 4 — ETC (2 marks): Inner mitochondrial membrane. Electrons from NADH and $FADH_2$ pass through carrier molecules releasing energy → ATP synthesis. NADH → 3 ATP each; $FADH_2$ → 2 ATP each. Final electron acceptor = oxygen → combines with H_2 → water. Total from ETC: 32 ATP. Final tally: Glycolysis 2 + Krebs 2 + ETC 32 = 36 ATP per glucose molecule.

IMPROVEMENT NOTED: The ETC location ("inner mitochondrial membrane") — which was missing in the April 30 attempt — is now explicitly stated. NADH:ATP and $FADH_2$:ATP ratios both given correctly. The 32 ATP from ETC is correctly derived, and the total 36 ATP breakdown (2+2+32) is written out. All four stages with locations, reactions, and yields present. Full 5/5.

MARKS SUMMARY

Section	Questions	Max Marks	Marks Obtained	%	Notes
Section A — MCQs	12 / 12 correct	12	**12**	100%	8th consecutive perfect MCQ score
Section B — Short (Q2)	All 11 attempted	33	**33**	100%	Every sub-part full marks
Section C — Long (Q3)	**All 4 attempted**	**20**	**20**	**100%**	**All 5-mark questions fully answered**
TOTAL		**65**	**65**	**100%**	**PERFECT — Second perfect score ever**

COMPARISON: This Attempt vs April 30 Attempt (Same Exam — Re-sit)

Biology exam-02-ch7 Head-to-Head

Metric	2026-04-30 (1st attempt)	2026-05-02 (Re-sit)	Change
Total Score	60.5/65	**65/65**	+4.5 marks
Percentage	93.1%	**100%**	+6.9%
Section A MCQs	12/12	12/12	—
Section B	29.5/33	**33/33**	+3.5 marks
Section C	19/20	**20/20**	+1 mark
Allosteric site named (Q2v)	Not named — 1 mark lost	Named explicitly	FIXED
ETC location named (Q2ix)	Not named — 1 mark lost	Inner mitochondrial membrane stated	FIXED
Enzyme specificity mechanism (Q2ii)	Missing active-site link — 1 mark lost	Full marks in cofactors question	FIXED
Photosynthesis equation (Q3iii)	Not written explicitly — 0.5 mark lost	Full marks in ATP question	FIXED

STRENGTHS AND AREAS TO WATCH

Strengths Demonstrated

- MCQ mastery — 8 consecutive perfect 12/12 scores (96/96 = 100% MCQ record now)
- All three previously identified weaknesses corrected in one sitting
- Vmax named explicitly for substrate concentration saturation (2-mark question)
- Pepsin pH 2.00 and Trypsin pH 8.00 correctly cited from memory
- Lock-and-Key (Emil Fischer, rigid) vs Induced-Fit (flexible, conformational change) distinction precisely drawn
- ATP-ADP cycle drawn as diagram with condensation/hydrolysis labelled
- All four aerobic respiration stages with correct locations and ATP breakdown (2+2+32=36)
- ETC: NADH→3 ATP, $FADH_2$→2 ATP ratios correctly given
- Tabular comparison for fermentation types — clean layout
- Complete coverage — no question left unanswered or incomplete

Points for Future Biology Exams

- No active weaknesses identified from this paper — all previous patterns corrected
- Watch for new chapter material: when studying Chapter 8 (Plant Physiology), apply the same precision — name organs/locations explicitly (e.g. phloem for translocation, xylem for water transport)
- When writing overall equations in long answers, always write the balanced equation explicitly even if it was stated in a short question earlier in the paper
- Continue the "mechanism behind facts" habit established here — this paper shows it is now consistent